

Numbness

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Numbness refers to the partial or complete loss of sensation. It can be a [symptom of nervous system malfunction](#).

People with numbness may be unable to feel light touch, pain, temperature, or vibration or to know where parts of their body are (position sense). When people do not know where parts of their body are, they have problems with balance and coordination.

Many people mistakenly use the term numbness when they have abnormal sensations such as tingling, prickling, or a pins-and-needles sensation or when a limb feels weak or is paralyzed—perhaps partly because people with numbness often also have such abnormal sensations and symptoms. The presence of other symptoms depends on what is causing numbness.

If numbness has been present a long time, particularly in the feet, it can lead to other problems. People may have difficulty walking and driving and may be more likely to fall. They may not notice infections, foot sores (ulcers), and injuries because they cannot sense pain as well. In such cases, treatment may be delayed.

Pathway for sensation

For a person to feel sensations normally, sensory receptors (specialized ends of sensory nerve fibers in the skin) must detect information in and around the body. These receptors must then send a signal along the following pathway:

- Through sensory nerves (nerves from the skin to the spinal cord)
- Through spinal nerve roots, formed by sensory nerves joined together into thick short branches that pass through the backbones (vertebrae) to connect with the spinal cord (see figure [How the Spinal Cord Is Organized](#))
- Up the spinal cord

- Through the brain stem
- To the part of the brain that perceives and interprets these signals (in the cerebrum)

For some parts of the body, the pathway includes a plexus or the cauda equina.

Plexuses are networks of sensory nerve fibers and motor nerve fibers (which carry signals from the brain and spinal cord to muscles and other body parts). In plexuses, these nerve fibers are combined and sorted to serve a particular area of the body. The fibers then branch off from the plexus to become peripheral nerves. There are 4 plexuses in the torso.

The **cauda equina** is a bundle of spinal nerve root fibers at the bottom of the spinal cord. This structure resembles a horse's tail, which is what its name means in Latin. It supplies sensation to the thighs, buttocks, genitals, and the area between them, which are called the saddle area because they are the area of the body that would touch a saddle.

Causes of Numbness

Numbness results when one part of the pathway for sensation malfunctions, usually because of a disorder or drug.

Many conditions can cause numbness in various ways. For example, they may:

- Reduce or block the blood supply to nerves in the body, as occurs in [vasculitis](#), or in the brain, as results from [stroke](#)
- Damage part of the pathway for sensation, as may result from [injuries](#) or from [hereditary disorders that affect nerves](#) (neuropathies), such as [Friedreich ataxia](#)
- Put pressure on (compress) part of the pathway
- Infect a nerve, as occurs in [leprosy](#), [HIV \(human immunodeficiency virus\) infection](#), or [Lyme disease](#)
- Cause nerves in part of the pathway to become inflamed and lose their outer layer (called demyelination), as occurs in [multiple sclerosis](#) or [Guillain-Barré syndrome](#)
- Cause metabolic abnormalities, which may occur in [diabetes](#), [vitamin B12 deficiency](#), or poisoning due to heavy metals (such as lead) or another toxin or when chemotherapy is used

Pressure on different parts of the pathway has various causes, depending on which part of the pathway is affected (see table [Some Causes and Features of Numbness](#)), as in the following:

- [Pressure on nerves](#): Repeating specific movements over and over, causing swelling, as occurs in [carpal tunnel syndrome](#), or remaining in one position too long, as when people sit with their legs crossed a long time
- Pressure on spinal nerve roots: [Rupture or herniation of a disk](#) in the spine, [osteoarthritis](#), or narrowing of the passageway for the spinal cord (spinal stenosis)
- [Pressure on the spinal cord](#): A tumor, an injury, or a pocket of blood (hematoma) or pus (abscess) near the spinal cord

Evaluation of Numbness

Because so many disorders can cause numbness, doctors ask questions systematically, focusing on more likely causes.

Warning signs

In people with numbness, the following symptoms are cause for concern:

- Numbness that begins suddenly (within minutes or hours)
- Weakness that accompanies the numbness and that begins suddenly or rapidly (within hours or days)
- Numbness or weakness that rapidly spreads up or down the body, involving more and more parts of the body
- Difficulty breathing
- Numbness in the thighs, buttocks, genitals, and the area between them (saddle area) and loss of bladder and bowel control (incontinence)
- Numbness on both sides below a specific level of the body (such as below the midchest)
- Numbness of an entire leg or arm
- Loss of sensation in the face and torso

When to see a doctor

People who have warning signs should go to a hospital immediately. People without warning signs should call their doctor. The doctor can decide how rapidly they need to be seen based on their symptoms.

What the doctor does

Doctors begin by asking which body parts are affected. The pattern of body parts affected by numbness often indicates which part of the nerve pathway is malfunctioning:

- Part of a limb: Peripheral nerve or sometimes spinal nerve root malfunction
- Arm and leg on the same side of the body: Brain malfunction
- Both sides of the body below a specific level of the body: Spinal cord malfunction, as occurs in [transverse myelitis](#) (which causes inflammation across the entire width of the spinal cord)
- Both sides, mainly in the hands and feet: Simultaneous malfunction of many peripheral nerves throughout the body (a [polyneuropathy](#))

Then doctors ask about the person's other symptoms and [medical history](#). Doctors also do a physical examination. What they find during the history and physical examination often suggests a cause and the tests that may need to be done (see table [Some Causes and Features of Numbness](#)).

Doctors ask the person to describe the numbness. Then doctors may ask specific questions:

- When numbness began
- How quickly it began
- Whether the person also has other symptoms such as abnormal sensations, weakness or paralysis, loss of bowel or bladder control, retention of urine, vision problems, difficulty swallowing, or deterioration of mental function
- Whether any event, such as pressure on a limb, an injury, sleeping in an awkward position, or an infection, triggered the symptoms

Knowing how quickly numbness and other symptoms began helps doctors determine the type of disorder.

The person is asked about symptoms that may suggest a cause. For example, back and/or neck pain suggests osteoarthritis, a ruptured disk, or another disorder that puts pressure on the spinal cord.

Doctors also ask whether the person has had a disorder that can cause numbness, particularly diabetes, chronic kidney disease, infections (such as HIV infection or Lyme disease), a stroke, or arthritis. Doctors may ask whether any family members have had

similar symptoms or have a hereditary disorder that affects the nervous system. They ask the person about use of medications and substances (including illicit drugs), and about possible exposure to toxins.

The physical examination includes a complete evaluation of the nervous system ([neurologic examination](#)), focusing on testing sensation (whether the person can feel stimuli, such as touch and temperature, normally), as well as reflexes and muscle function.



Some Causes and Features of Numbness

Cause	Common Features*	Diagnostic Approach†
Numbness in both limbs (arm and leg) on one side of the body		
Disorders that affect the cerebrum (the largest area of the brain), such as	Loss of sensation on the opposite side of the face and body from the side of the brain that is affected by the disorder and loss of the ability to recognize items by touch	MRI or CT of the brain
• Stroke • Tumors • Multiple sclerosis • Degenerative brain disorders	Usually weakness, loss of coordination, and other symptoms indicating malfunction of the nervous system	
Disorders that affect the upper part of the brain stem, such as	Loss of sensation on the opposite side of the face and body from the side that is affected by the disorder	MRI or CT of the brain
• Stroke • Tumors • Abscesses (pockets of pus)	Often double vision	
Disorders that affect the lower part of the brain stem, such as	Loss of sensation on same side of the face and on the opposite side of the body from the side that is affected by the disorder	MRI of the brain
• Stroke • Tumors • Degenerative brain disorders	Often vision problems and difficulty chewing,	



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